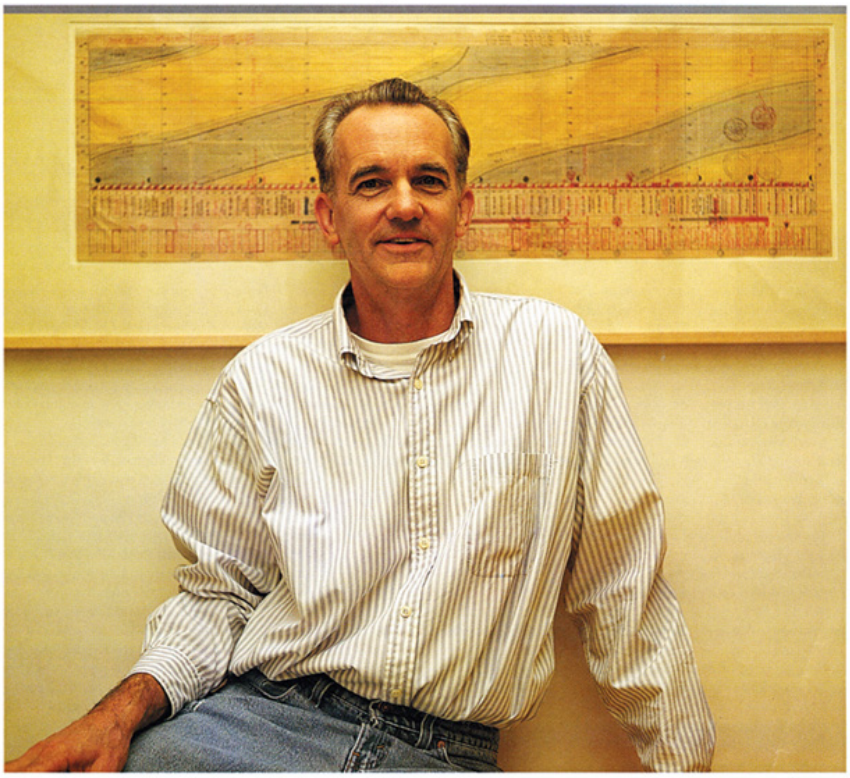
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Fortune Magazine, October 27, 1997

SMART MANAGING

BEST PRACTICES, CAREERS, AND IDEAS



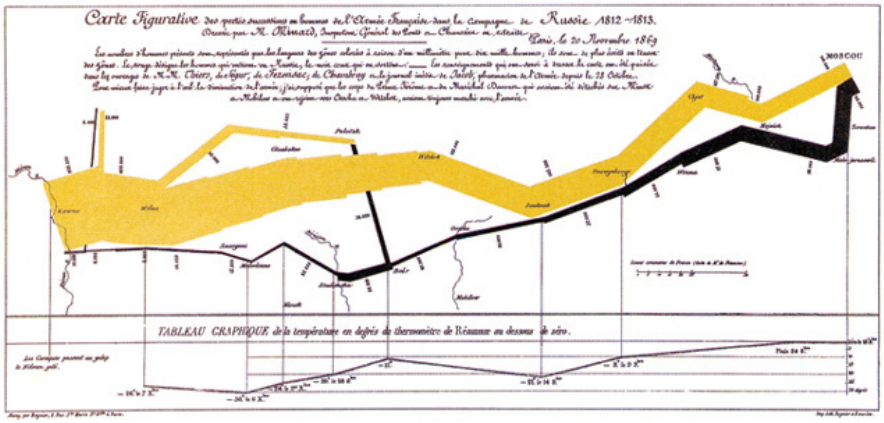
The Man Who Makes Sense of Numbers

Yale professor Edward Tufte dazzles business people by making rational the data that rule their work lives.

by Michael H. Martin

Do you remember, perhaps from a college English course, a skinny little book called *The Elements of Style*. Writers consider it a gem among reference books, a practical guide that gracefully explains the principles of good writing. Turns out that there's a similar guide for those who work with numbers for a living—which these days means just about everyone. The book is *The Visual Display of Quantitative Information*. Written by Edward Tufte, a statistician, designer, and Yale professor, that book and its two sequels—*Envisioning Information* and *Visual Explanations*—have become bibles of design for thousands of engineers, computer designers, scientists, and financial analysts.

Tufte, 55, a lanky, precise man who dresses a bit better than most academics, showed few signs of such eminence early in his career. After receiving degrees in statistics and political science from Stanford and Yale, he took a teaching job at Princeton. His course work there eventually led to his first book, which he self-published us-



ing funds from a second mortgage on his home. ("Nothing like an 18% interest rate to concentrate the mind," he says.) He had 5,000 copies printed up, rented a post office box, and placed an ad in *Scientific American*. The last of his cash paid for 2,400 of the 5,000 copies. Then he and his wife went away for a few days. "The mailbox filled up just like that," he says. "We came back Saturday morning, and it was stuffed." Since then the money and accolades have kept rolling in. Although he eventually hired several people to help with marketing, shipping, and logistics, Tufte still ships most of the books out of his garage.

With some 460,000 copies in print, the books have earned Tufte not only millions of dollars but a position of rare influence as well. You might compare his works with Paul Samuelson's famous economics textbook, which established Samuelson's Keynesian views as the standard among generations of undergraduates. Tufte's books have something of the same status among numbers folks. They have also generated an amazingly strong mainstream interest.

Tufte fills hotel ballrooms around the country with business people eager to hear his sharp critiques and thoughts on the "deep principles of information design." Tufte says he wants to "change the way people see," though most attendees have more utilitarian aims. Says one longtime Tufte fan: "People who follow the principles he talks about can stop reading about themselves in *Dilbert*."

Among the 400 or so in the crowd at a recent talk in San Francisco were any number of software programmers and management-consulting types, plus a couple of dozen managers from Barclay's Global Investors, several sales reps from Compaq and Apple Computer, a procurement expert for the U.S. Navy, and a nuclear weapons physicist from Lawrence

In his books, Tufte points to inspired designs that tell a rich, coherent story—such as this 1869 map portraying Napoleon's escalating losses in the Russian campaign of 1812.

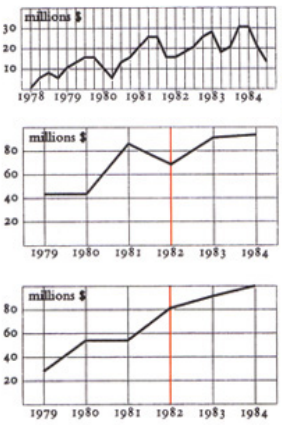
National Laboratories—not to mention a flock of multiply pierced and tattooed Web-design hipsters. Each member of this motley crowd paid $300 to listen to Tufte speak about the art and science of visual devices—from the cave drawings of France, to early astronomers' attempts to render sunspots, to supercomputer animation of a thunder cloud.

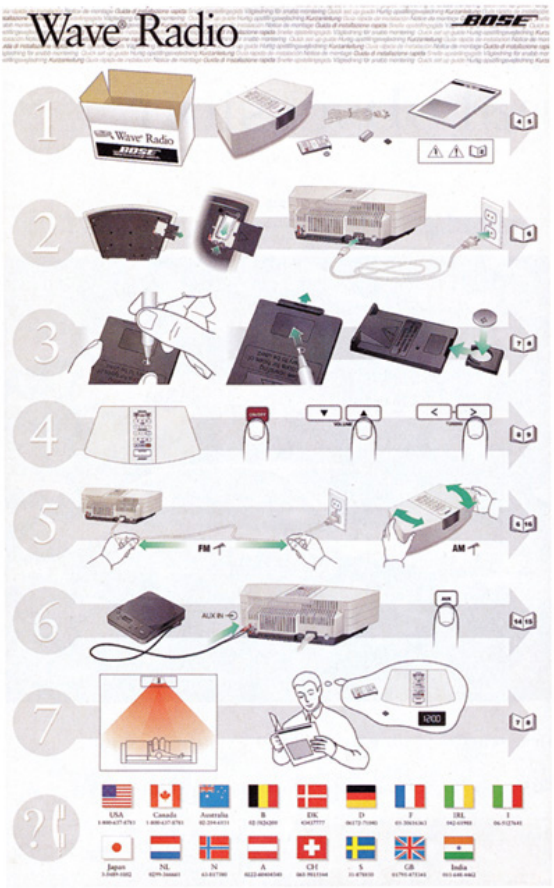
Tufte's core insight is that clear and intelligent design not only reflects but actually spurs clear and intelligent thought. This notion has obvious implications for designers and engineers, but it also applies to business people faced with making decisions on the basis of charts and graphs. "You have to be careful, because the exhibits in front of you can set the agenda," says Tufte.

When he says something like, "If the numbers are boring, then you've got the wrong numbers," he points beyond the accuracy of the data and toward the ways that inspired design can rivet attention on the issues the data illustrate. Actually, what Tufte says is both more intelligent and more abstruse: "Clutter and confusion are failures of design, not attributes of information." To illustrate this insight, Tufte mercilessly critiques the ways information is usually presented to us on the page or on the screen. Bad graphics, says Tufte, fail because they omit or manipulate context, deceive by discouraging comparison or obscuring important details, and confuse with visual miscues. We are, he says, surrounded by poorly rendered data: annual reports that portray years of rising revenues without adjusting for inflation; the Surgeon General's cigarette warning undercut by its heavy black border and indistinguishable letters.

Such lapses are more than mere aesthetic offenses. Confusing hospital charts lead to improper diagnosis and treatment, for example. Tufte's most remarkable illustration of the link between visual and

A Tufte primer on data manipulation: The top chart shows quarterly revenue, with several dips. Middle, the same data by fiscal year produces one dip. Bottom, the dip is gone if the data are shown by calendar year.





Tufte helped develop this poster, which explains how to set up a home audio system, for a Massachusetts company. Since there's no text, it can be used all over the world.

But he's rather sour on the experience: "The time was never right for a professor to come in and make suggestions.... I think it's something they learn in product manager school."

It's easy for an outsider to make such criticisms, of course. But even Tufte's cheap shots are accompanied by some remarkable insights. Almost in passing, Tufte points out that most corporate Websites "mimic the structure of the bureaucracy producing the design rather than complement the information presented." (Think about that and try surfing the Web. Suddenly all those overproduced, underfunctional Websites make an eerie sense.)

Tufte has had more success with other clients, especially those with products whose success or failure is largely determined by design issues. At Bose Corp., a Framingham, Mass., company that makes high-end audio products for audiophiles and professional venues like stadiums and recording studios, Tufte has become something of a guru. "People at Bose use his name as a verb," says Ken Jacob, Bose's director and chief engineer of professional projects. "They'll ask, 'Have you Tufte-ized that chart yet?' "

At Bose, Tufte suggested improvements on everything from product manuals to the financial portrait presented to the company's directors. He helped create a poster, explaining how to set up complicated audio systems, that doesn't rely on text. Now Bose uses the same poster anywhere in the world—no translation necessary. And when Jacob asked Tufte to critique the portfolios that Bose salespeople use to track big clients, Tufte surprised him: "I expected him to comment on the quality of the charts. What he said was totally unexpected. He asked how much it costs to measure all these things, and is it worth it? They had collected a very info-rich data set and had done a good job of design.... But he caught them on an idea: whether what you measured is worth it."

statistical thinking is his analysis, in *Visual Explanations,* of the process that led to the disastrous flight of the space shuttle *Challenger* in 1986. With heartbreaking thoroughness, Tufte lays out the memos, charts, and tables that were sent from the aerospace engineers to NASA in the 12 hours before they decided to go ahead with the launch. He demonstrates how one simple graph of the data they had at hand—information about the failure of the booster rocket's O-rings at various temperatures—would have alerted them to the danger they faced.

Tufte is particularly cynical about the way computers have affected the display of information. (A favorite saying: "There is only one other industry that calls the people who buy its products 'users.' ") In fact, Tufte has consulted for lots of computer companies, including IBM, Hewlett-Packard, and Sun Microsystems.

Tufte's books provide step-by-step accounts of how to edit bad information displays, erasing distractions and replacing meaningless labels with self-explanatory ones. They also provide an archive of inspiring examples, ranging from 1,000-year-old Chinese maps, to historical graphs of Napoleon's march on Moscow, to a touch-screen information kiosk.

Most important are the principles that he sets forth: Tell the truth. Show the data in its full complexity. Reveal what is hidden. Especially, respect the reader: Tufte vehemently denounces the presumption that graphics are "devices for showing the obvious to the ignorant." It's an irony of the modern age that in a world offering so many sources of data, we are often no more accurately informed than our forebears. Tufte's principles offer solutions for that state of affairs, and not a moment too soon. P

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